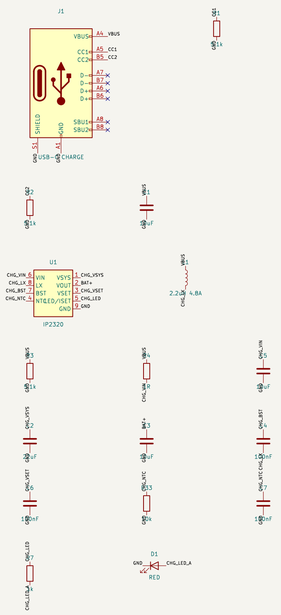
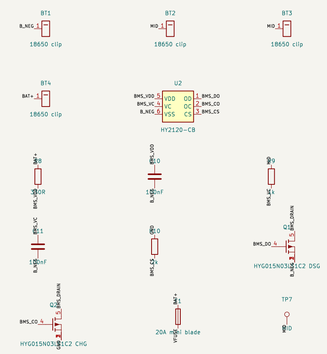
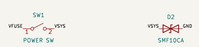
1 USB-C charge input + IP2320 2S boost charger



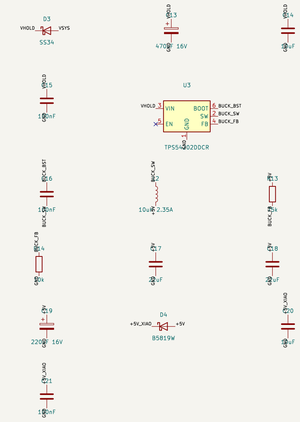
2 Cells, 2S BMS (HY2120), 20 A blade fuse



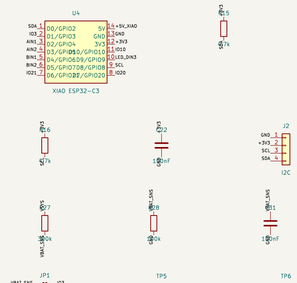
3 Power switch (off-board), surge clamp, bulk



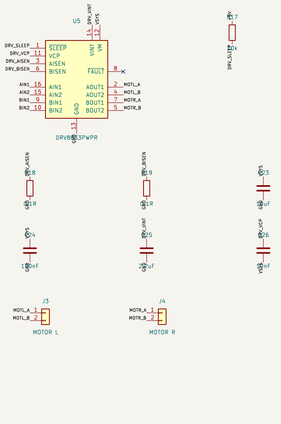
4 Hold-up + 5 V buck (TPS54202), XIAO feed



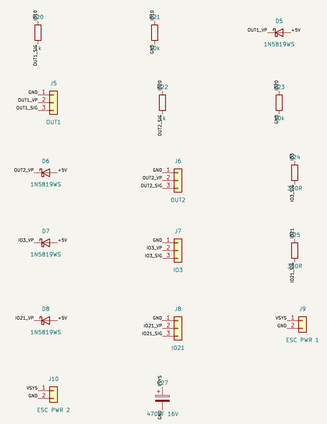
5 XIAO ESP32-C3, I2C, battery sense



6 DRV8833 drive, 2 A current limit



7 PWM outputs (ESC or servo) + spare GPIO



8 RGB status LED (WS2812B strip piece)



Power flags (ERC)

#FLG01 PWR_FLAG
#FLG02 PWR_FLAG
#FLG03 PWR_FLAG
#FLG04 PWR_FLAG
#FLG05 PWR_FLAG
#FLG06 PWR_FLAG
#FLG07 PWR_FLAG
#FLG08 PWR_FLAG
#FLG09 PWR_FLAG
#FLG010 PWR_FLAG
#FLG011 PWR_FLAG
#FLG012 PWR_FLAG
#FLG013 PWR_FLAG
#FLG014 PWR_FLAG
#FLG015 PWR_FLAG
#FLG016 PWR_FLAG

Cardboard Bot v2 main board — rev A — Columbia Gadget Works
Generated from hardware/gen/circuit.py (do not hand-edit; edit circuit.py and regenerate).
All connections are made by net labels at pin ends.
XIAO pins: D0 SDA(GPIO2) D1 IO3 D2–D5 DRV8833 AIN1/AIN2/BIN1/BIN2 D6 IO21 D7 OUT2(GPIO20) D8 SCL(GPIO8) D9 LED(GPIO9) D10 OUT1(GPIO10)
IP2320 per INJOINIC V1.00 datasheet fig. 5; HY2213 per HYCON DS-HY2213-V05 P-MOSFET example; clip per MYOUNG WD-CP-0056.
Battery sense: VSYS/4 on GPIO3 through bridged jumper JP1 (cut JP1 to use IO3 as plain GPIO).

Columbia Gadget Works
Sheet: /
File: cardboard-bot.kicad_sch
Title: Cardboard Bot v2 main board
Size: A1 | Date: 2026-10-27 | Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 1/1